FreqODES: Frequency-Domain Guided Online Domain Adaptation with Expert-in-the-Loop Active Learning

Replication and extension of ODES (MICCAI 2025)

Group <No.>

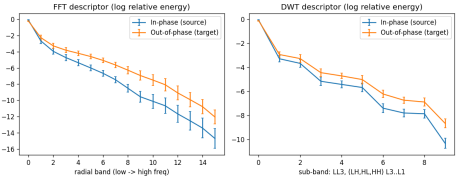
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22AIE437 Medical Image Processing
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Problem and motivation

- Abdominal organ segmentation (liver, kidneys, spleen) from MRI supports surgical planning, volumetry and follow-up.
- A network trained on one scanner / sequence (*source*) degrades on another (*target*): **domain shift**.
- Example: CHAOS T1-DUAL **in-phase** $\rightarrow$ **out-of-phase**: same patients, same organs, yet Dice of our source model drops from training level to **54.97%**.
- Retraining per hospital needs costly expert labels; unsupervised test-time adaptation relies on noisy pseudo-labels.
- **Online active domain adaptation**: adapt patient-by-patient, asking an expert to label only a tiny budget of pixels.



Same anatomy, different MR sequence: OOP has more high-frequency energy (FFT bands, left; DWT sub-bands, right).

Dataset, task and objectives

Dataset: CHAOS challenge [2] (MR, T1-DUAL)

- 20 patients, 647 axial slices (26–50 per patient), 256×256 / 288×288, resized to 256^2
- Labels: liver, right kidney, left kidney, spleen (+ background) = 5 classes
- **Source**: in-phase (IP), all 20 volumes, used for training
- **Target**: out-of-phase (OOP), the 20 volumes arrive as an *online stream*, one patient = one batch (paper protocol, no train/test split on target)

Task: 2-D multi-class **segmentation**

Objectives

1. Reproduce ODES [1] on CHAOS IP→OOP.
2. Replace its BN-statistics image-pruning score with an interpretable **FFT / DWT frequency-domain** shift score (FreqODES).
3. Compare at equal annotation budget ($K = 10, 50, 100\%$ slices, $b = 1\%$ pixels).

Metrics: Dice (DSC), Jaccard (IoU) per organ and mean; adaptation time; paired Wilcoxon test over the 20 patients.

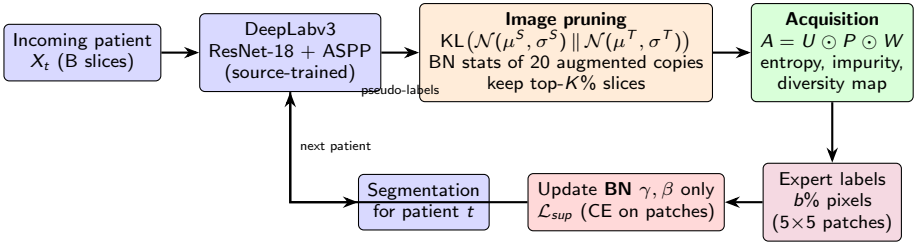
Base paper

ODES: Online Domain Adaptation with Expert Guidance for Medical Image Segmentation

M. S. Islam, S. Nag, A. Dutta, S. M. Ahmed, F. F. Niloy, S. Bera, A. K. Roy-Chowdhury
University of California, Riverside — **MICCAI 2025** [1]. Code: `github.com/ShazidAraf/ODES`

- First active-learning based *online* UDA framework for medical segmentation: source-free, storage-free, one pass per batch.
- Four stages per incoming patient: **infer** $\rightarrow$ **prune images** $\rightarrow$ **acquire patch labels** $\rightarrow$ **update BN layers**.
- Contributions: (i) **image pruning** by BN-statistics KL divergence; (ii) **diversity-aware acquisition** (entropy $\times$ regional impurity $\times$ farthest-first diversity map).
- Evaluated on CHAOS IP$\rightarrow$OOP, CHAOS$\rightarrow$DUKE, BMC$\rightarrow$RUNMC; beats TENT, CoTTA, SaTTCA, ...

ODES architecture



- **Backbone**: DeepLabv3 [3], ResNet-18 encoder (ImageNet init), ASPP, low-level skip, 5-class head.
- **Pruning**: each slice is augmented (flip / rotation), passed through a copy of the net; BN statistics of the last 9 BN layers are compared with source running statistics (Eq. 1 of [1]).
- **Acquisition**: U = patch-mean entropy, P = regional impurity [4], W = Gaussian map around farthest-first-traversal picks (spatial $\times$ feature).

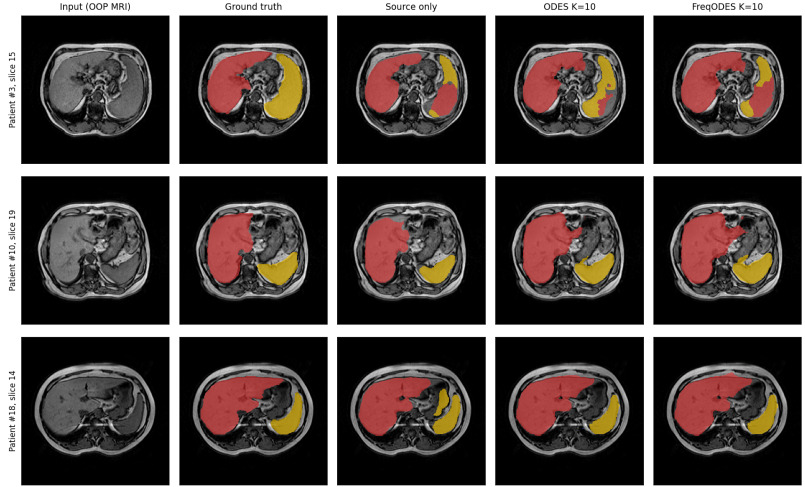
Replicated vs. reported results (CHAOS IP $\rightarrow$ OOP, $b = 1\%$)

Table 1: Dice (%) per organ; Jaccard (IoU, %) mean. Paper values from Table 1 of [1].

	Setting	Liver	L. Kidney	R. Kidney	Spleen	**Mean DSC**	**Mean IoU**
Source only	Paper	87.77	37.97	18.92	67.31	52.99	–
	Ours	85.89	29.84	33.74	70.43	54.97	42.74
ODES K=100	Paper	88.90	72.41	74.55	78.56	78.61	–
	Ours	87.89	76.90	75.28	78.00	**79.52**	66.96
ODES K=50	Paper	89.13	72.30	74.43	78.36	78.56	–
	Ours	88.02	76.66	75.71	77.89	**79.57**	67.00
ODES K=10	Paper	88.41	71.45	70.22	77.12	76.80	–
	Ours	87.88	75.19	74.18	78.06	**78.83**	66.24

- Source-only and adapted results are within $\approx$2 DSC of the paper; the adaptation gain (54.97$\rightarrow$79.52, $\approx +25$) and the K trend ($100 \approx 50 > 10$) are reproduced.
- No source checkpoint or training recipe is released, so we trained our own source model; organ-level differences (e.g. kidneys) come from that.
- Paper reports no Jaccard; ours is measured per volume.

Qualitative results: input, ground truth, predicted masks



Red: liver, green: right kidney, blue: left kidney, yellow: spleen. Three target (OOP) patients, slice with largest organ area.

Code structure and training settings

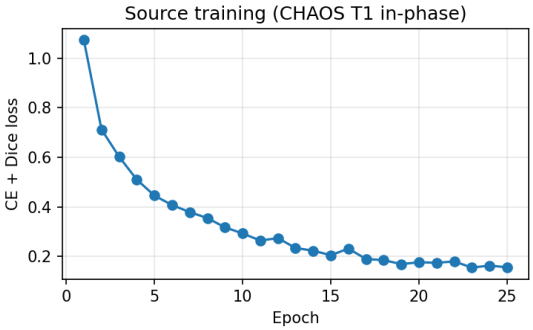
```
prepare_chaos.py    DICOM/PNG -> (D,H,W) tif stacks
dataloader/         Stack_Loader: one patient = batch
model/              DeepLabv3 (ResNet-18 + ASPP)
train_source.py     source training (CE + Dice)
methods/ODES.py     pruning, acquisition, BN update
methods/freq_shift.py  NEW: FFT / DWT shift score
FFT_weight_2.py     diversity map (missing upstream,
                    re-implemented from the paper)
demo_ODES.py        online loop + Dice / IoU eval
run_all.sh          21-run experiment sweep
make_report.py      tables, plots, LaTeX numbers
```

Upstream repo had no training script, no checkpoint and a missing module; we added these and fixed CPU/GPU and a stale-variable bug.

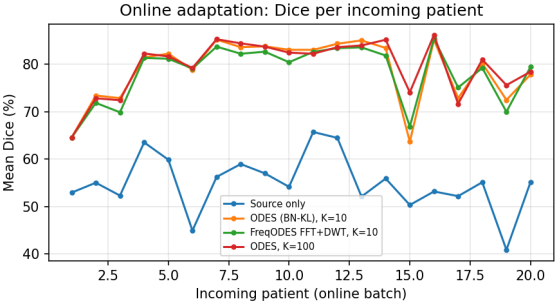
Table 2: Training / adaptation settings

Framework	PyTorch 2, timm
Source optimiser	Adam, lr 10^{-3}, wd 10^{-4}, cosine
Batch size / epochs	8 / 25
Loss	CE + Dice
Augmentation	flips, 90° rot., intensity jitter
Adaptation optimiser	Adam, lr 10^{-2}, 1 step / patient
Updated params	BN γ, β only
Budget	$b = 1\%$ px, $K \in \{10, 50, 100\}\%$
Hardware	NVIDIA Tesla T4 (Colab)
Seed	1337 (all runs)

Training log and curves



Source training loss (25 epochs, 647 IP slices): $1.07 \rightarrow 0.16$.



Online adaptation: mean Dice of each incoming target patient.

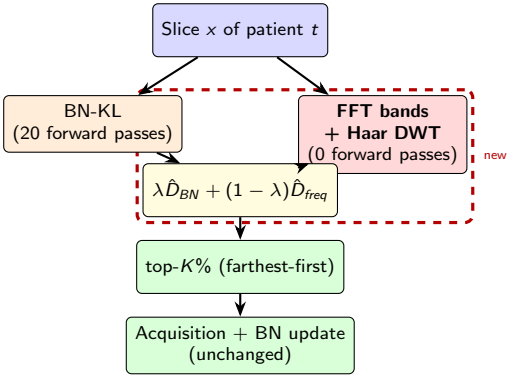
Expert-guided BN updates lift every patient above source-only.

What we changed: frequency-domain image pruning

Hypothesis: MR sequence / scanner shift is directly visible as a change in spatial-frequency energy, so it can be detected from the image itself, without the network.

- **FFT descriptor**: log fraction of energy in 16 concentric radial bands of $|\mathcal{F}(x)|^2$.
- **DWT descriptor**: log relative energy of LL_3 and LH/HL/HH sub-bands of a 3-level Haar DWT.
- Reference μ_f, σ_f from 200 random source slices.
- Shift score $D_{freq}(x) = \frac{1}{d}\sum_f \left(\frac{\phi_f(x)-\mu_f}{\sigma_f}\right)^2$
- **Fused**: $D = \lambda \hat{D}_{BN} + (1-\lambda)\hat{D}_{freq}$

Only Stage 3 changes; backbone, acquisition, update and budgets are identical $\Rightarrow$ fair comparison.



Does the frequency score detect the shift?

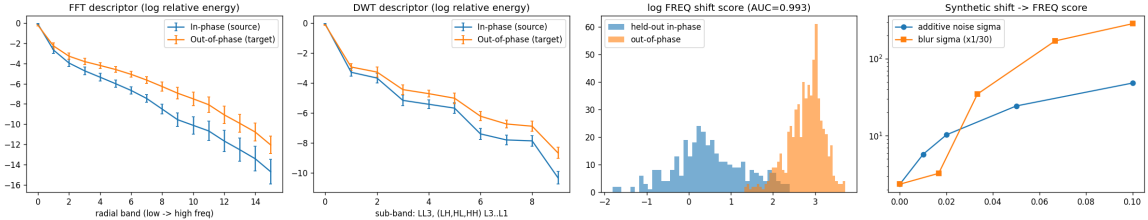


Table 3: Separating held-out IP slices (10 patients) from OOP slices with the shift score alone.

Descriptor	FFT	DWT	FFT+DWT
ROC-AUC	**0.994**	0.969	0.993

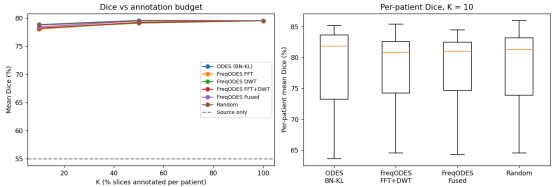
OOP slices carry systematically more high-frequency energy; the score also grows monotonically with synthetic noise and blur.

FreqODES v1 (frequency score alone) vs. ODES, single seed

Table 4: Same data, model, seed and budget ($b = 1\%$). DSC / IoU in %, time = full online pass over 20 patients on a T4. p: paired Wilcoxon vs. ODES at the same K.

	$K = 10\%$ of slices				$K = 50\%$ of slices			
Pruning score	DSC	IoU	p	Time (s)	DSC	IoU	p	Time (s)
BN-KL (ODES, replicated)	**78.83**	66.24	–	167	79.57	67.00	–	171
FFT (FreqODES)	78.02	65.03	0.005	107	79.24	66.61	0.064	115
DWT (FreqODES)	78.15	65.28	0.027	109	79.09	66.43	0.030	114
FFT+DWT (FreqODES)	78.20	65.27	0.091	109	79.20	66.52	0.044	114
Fused, λ=0.5	78.40	65.45	0.159	169	**79.50**	66.92	0.687	173
Random	78.75	66.04	0.334	107	79.52	67.00	1.000	113

All slices ($K = 100$): DSC 79.52, IoU 66.96 Source only: DSC 54.97, IoU 42.74



Ablations: pruning $\times$ diversity weighting, and fusion weight

Table 5: Mean DSC at $K = 10$ (cf. Table 3 of [1]).

Image pruning	D.W. off	D.W. on
BN-KL (ODES)	78.73	78.83
FFT+DWT (ours)	78.32	78.20
Random	78.96	78.75
Paper (BN / random)	75.88 / 74.32	76.80 / 75.41

Table 6: Fusion weight λ ($K = 10$).

λ	Mean DSC
0 (pure frequency)	78.20
0.25	79.09
0.5	78.40
0.75	78.93
1 (pure BN = ODES)	78.83

Mixing a little BN information into the frequency score ($\lambda = 0.25$) gives the best $K = 10$ result (79.09 vs. 78.83, not significant, $p = 0.75$); diversity weighting changes DSC by ≤ 0.2 in our runs, much less than reported in the paper.

Analysis: why v1 alone did not beat ODES (and what it led to)

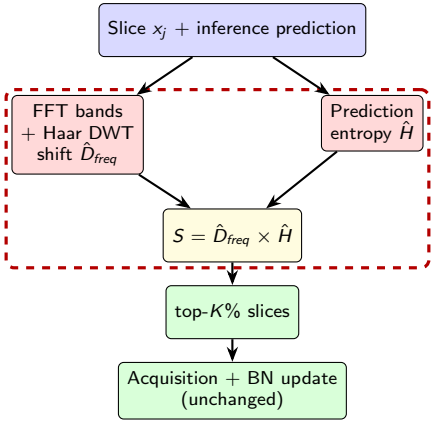
- **Where it helps**: best shift *detector* (AUC 0.994), interpretable, needs no forward passes $\Rightarrow$ full online pass 109 s vs. 167 s ($\approx$ 35% faster). The fused score matches ODES at both budgets (79.50 vs. 79.57 at $K = 50$, $p = 0.687$) and with $\lambda = 0.25$ reaches 79.09 at $K = 10$.
- **Where it hurts**: pure frequency scores trail ODES by 0.6–0.8 DSC at $K = 10$ (FFT: -0.81, $p = 0.005$) and by 0.3–0.5 at $K = 50$.
- **Mismatch between objective and score**: the most frequency-shifted slices are the volume extremes (few or tiny organs, mostly background); labelling them teaches the BN layers little. BN-KL measures what *the network* finds unusual, which correlates better with segmentation error.
- **Small, homogeneous shift**: IP$\rightarrow$OOP shift is global, so nearly every OOP slice is "shifted"; ranking among them is what matters, and random pruning is already strong (78.75, $p = 0.334$).
- **Fix**: a slice is worth labelling if it is *shifted* **and** the model is *unsure* about it $\Rightarrow$ multiply the frequency score by model uncertainty (next slide), and evaluate over 3 seeds.

FreqODES-U: frequency shift × model uncertainty

Slice score

$$S(x_j) = \underbrace{\hat{D}_{freq}(x_j)}_{\text{FFT+DWT shift}} \times \underbrace{\hat{H}(x_j)}_{\text{mean pixel entropy}}$$

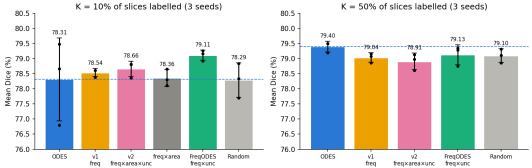
- $\hat{D}_{freq}$: v1 frequency score; $\hat{H}$: mean entropy of the softmax prediction of the slice; each normalised by its batch maximum.
- $\hat{H}$ reuses the inference prediction ODES already makes ⇒ still **0 extra forward passes**.
- Pre-registered design (v2): $\hat{D}_{freq} \times$ organ area $\times \hat{H}$, with two ablations (drop one factor each). The ablation $\hat{D}_{freq} \times \hat{H}$ was best; we report *all* variants.
- Protocol: 3 seeds (1, 2, 12345) × 20 patients; paired Wilcoxon over the 60 patient-runs.



FreqODES-U results: 3 seeds, same model and budget

Table 7: Mean Dice (%) ± std over 3 seeds; Δ and p: paired Wilcoxon vs. ODES over 60 patient-runs; time: full online pass (T4).

	$K = 10\%$ of slices				$K = 50\%$ of slices		
Selection score	DSC	Δ / p	IoU	Time	DSC	Δ / p	IoU
ODES (BN-KL)	78.31±1.38	–	65.51	154 s	**79.40**±0.18	–	66.90
v1: frequency	78.54±0.13	+0.22 / 0.908	65.63	104 s	79.04±0.15	-0.36 / 0.000	66.36
v2: freq × area × unc.	78.66±0.26	+0.35 / 0.199	65.84	105 s	78.91±0.27	-0.49 / 0.000	66.18
Ablation: freq × area	78.36±0.28	+0.04 / 0.700	65.57	104 s	–	–	–
FreqODES-U: freq × unc.	**79.11**±0.17	**+0.80 / 0.004**	**66.43**	104 s	79.13±0.33	-0.27 / 0.018	66.44
Random	78.29±0.57	-0.02 / 0.232	65.40	102 s	79.10±0.23	-0.30 / 0.001	66.43



At $K = 10\%$ FreqODES-U beats ODES on every seed (78.93, 79.16, 79.26 vs. 76.80, 78.67, 79.48), 8× more stable, ≈33% faster. At $K = 50\%$ ODES stays best.

Conclusion, contributions and references

Conclusion

- ODES reproduced on CHAOS IP→OOP (79.52 vs. 78.61 DSC at $K = 100$) after completing the public code.
- v1: an interpretable FFT/DWT shift detector (AUC 0.994), $\approx$ 35% cheaper, but alone it does not beat ODES: a good *shift* detector is not automatically a good *annotation-value* detector.
- **FreqODES-U** (frequency $\times$ uncertainty): **79.11 vs. 78.31 DSC at** $K = 10\%$ (Δ+0.80, $p = 0.004$, 3 seeds), more stable and $\approx$33% faster; at $K = 50\%$ ODES stays best (79.40 vs. 79.13).
- Caveats: best variant found in a pre-planned ablation; one dataset pair; 3 seeds. Next: CHAOS→DUKE, more seeds.

Member contributions

- Member 1 Name (Roll No.): <contribution>
- Member 2 Name (Roll No.): <contribution>

[1] M. S. Islam, S. Nag, A. Dutta, S. M. Ahmed, F. F. Niloy, S. Bera, A. K. Roy-Chowdhury. ODES: Online domain adaptation with expert guidance for medical image segmentation. *MICCAI*, 2025.

[2] A. E. Kavur et al. CHAOS challenge – combined (CT-MR) healthy abdominal organ segmentation. *Medical Image Analysis* 69:101950, 2021.

[3] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, A. L. Yuille. DeepLab: semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs. *IEEE TPAMI* 40(4):834–848, 2018.

[4] B. Xie, L. Yuan, S. Li, C. H. Liu, X. Cheng. Towards fewer annotations: active learning via region impurity and prediction uncertainty for domain adaptive semantic segmentation. *CVPR*, 2022.